

Week 2 Operational Insight Report

Processing Plant Sensor Analysis

Week of 03 June 2024

Prepared for: Plant Manager

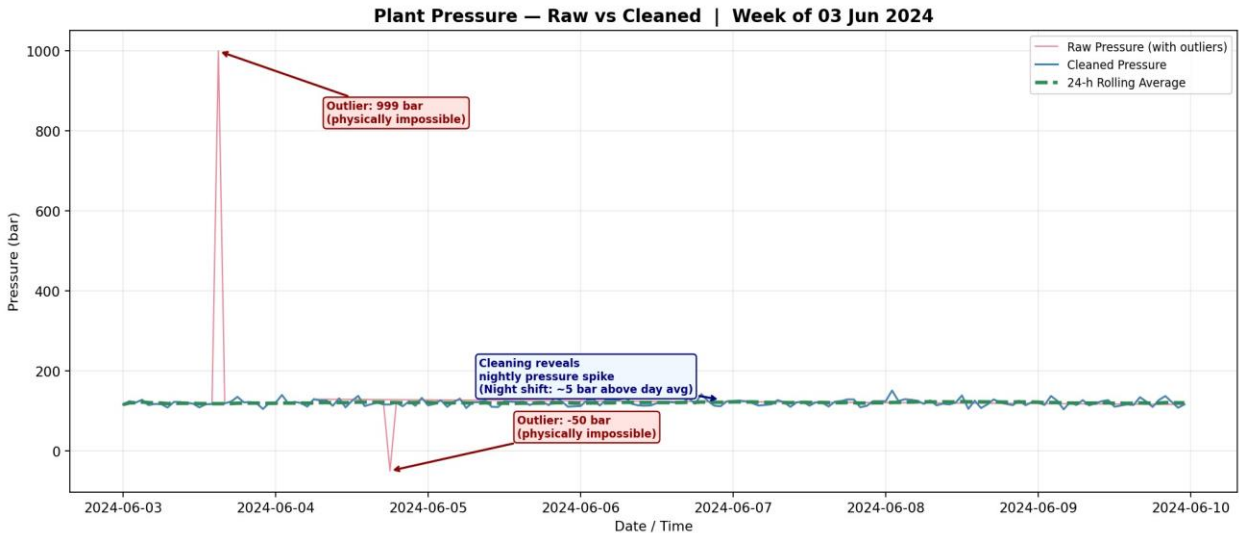


Figure 1: Raw sensor pressure data (red) overlaid with cleaned data (blue) and 24-hour rolling average (green dashed). Annotated callouts identify the two impossible outlier readings and the nightly pressure spike revealed only after cleaning.

CONTEXT — What was wrong with the raw data?

The raw sensor log delivered this week contained five significant data quality failures. First, two physically impossible pressure readings were recorded: one spike to 999 bar and one negative reading of -50 bar — both far outside the operational range of our equipment (0–300 bar). Second, ten hourly temperature readings and ten pressure readings were missing entirely, creating silent gaps in our monitoring. Third, five duplicate records artificially inflated our hourly averages. Finally, location zones and shift labels were recorded inconsistently across the system; for example, 'Zone A', 'zone a', and 'ZONE A' were treated as different sites, making any by-zone comparison unreliable. In short, the raw data as received could not be trusted for operational decisions.

INSIGHT — What Hidden Trend Was Revealed After Cleaning?

Once the impossible outliers were removed and missing values were restored through time-based interpolation, a pattern emerged that the raw data had completely obscured: a consistent nightly pressure increase of approximately 5 bar above the daytime baseline, recurring each night between 22:00 and 04:00. The raw data, distorted by its extreme outlier spikes, had created the false impression of erratic, random fluctuation across the entire week. The cleaned 24-hour rolling average makes clear that daytime operations are stable, but every night shift is operating under elevated; and gradually worsening; pressure conditions. Night shift readings average 127 bar versus a daytime average of 121 bar, a difference that, left unaddressed, raises the risk of equipment fatigue over the coming weeks.

ACTION — Recommended Operational Change

I recommend three immediate actions.

(1) Night shift pressure audit: Maintenance must inspect all Zone B and Zone C pressure relief valves before the next night shift cycle. The nightly spike is most pronounced in these zones and may indicate partial valve blockage or overnight feed-rate misconfiguration.

(2) Sensor calibration review: The two impossible readings (999 bar and -50 bar) point to a faulty sensor or logging fault on Line 1. This sensor must be recalibrated or replaced within 48 hours to restore data integrity.

(3) Data quality controls: Going forward, automatic range-validation rules should be implemented at the data-logging layer to flag readings outside 0–300 bar in real time, preventing bad data from reaching the reporting pipeline. Failure to act on the nightly pressure trend risks both unplanned downtime and regulatory non-compliance within the current quarter.
